CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 09

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In science, a 'law' usually: 1

(a) explains why the pattern occurs in every case
(b) describes a regular pattern observed in nature
(c) is a guess without evidence
(d) can never be tested

2. Measurements in science are based on agreed international standards rather than: 1

(a) units **(b)** instruments
(c) local objects or opinions **(d)** numbers

3. Which of the following is a homogeneous mixture? 1

(a) sand in water **(b)** smoke in air
(c) chalk powder in water **(d)** soda water (CO_2 in water)

4. Slow cooling of a hot saturated solution gives: 1

(a) no crystals **(b)** very small crystals
(c) larger, well-formed crystals **(d)** a gas

5. The molecular/formula unit mass of magnesium chloride, $MgCl_2$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 190 u **(b)** 93 u **(c)** 95 u **(d)** 97 u

6. Water always contains hydrogen and oxygen in the mass ratio: 1

(a) 2 : 1 **(b)** 8 : 1 **(c)** 1 : 8 **(d)** 1 : 1

7. The electronic configuration (K, L, M, N shells) of the calcium atom (Z = 20) is: 1

(a) 2, 8, 9, 1 **(b)** 2, 8, 8, 3 **(c)** 2, 9, 9 **(d)** 2, 8, 8, 2

8. A body starts with velocity 10 m s^{-1} and accelerates uniformly at 2 m s^{-2} for 4 s. The distance travelled is: 1

(a) 32 m **(b)** 48 m **(c)** 56 m **(d)** 112 m

9. The slope of a position–time graph gives: 1

(a) velocity **(b)** distance only **(c)** displacement **(d)** acceleration

10. A net force of 12 N acts on a body of mass 8 kg. Its acceleration is: 1

(a) 96 m s^{-2} **(b)** 3 m s^{-2} **(c)** 1.5 m s^{-2} **(d)** 0.67 m s^{-2}

11. For the same force, a body with larger mass has: 1

(a) the same acceleration **(b)** larger acceleration
(c) smaller acceleration **(d)** zero acceleration

12. A crowbar is a lever of the class in which: 1

(a) there is no fulcrum **(b)** the load is between fulcrum and effort
(c) the effort is between fulcrum and load **(d)** the fulcrum is between effort and load

13. The quality of sound that lets us distinguish a sitar from a flute playing the same note is: 1

(a) pitch **(b)** amplitude **(c)** echo **(d)** timbre

14. Genetically regulated cell death is called: 1

(a) programmed cell death **(b)** plasmolysis
(c) cancer **(d)** mitosis

15. If 25 cells lie along a 5 mm diameter of the field of view, the size of one cell is: 1

(a) 0.5 mm **(b)** 0.02 mm **(c)** 0.2 mm (200 µm) **(d)** 2 mm

16. The tissue that lines the lungs' air sacs and is very thin is: 1

(a) columnar epithelium **(b)** cuboidal epithelium
(c) squamous epithelium **(d)** ciliated muscle

17. Bacteria belong to the kingdom: 1

(a) Fungi **(b)** Monera **(c)** Plantae **(d)** Protista

18. Ovules are present in the: 1

(a) anther **(b)** stigma **(c)** ovary **(d)** style

19. The experiments of which scientist proved that life comes from pre-existing life? 1

(a) Whittaker **(b)** Mendel **(c)** Hooke **(d)** Louis Pasteur

20. If the insolation is 1 kW m^{-2}, the solar energy received by 5 m^2 in 2 h is: 1

(a) 7 kWh **(b)** 20 kWh **(c)** 10 kWh **(d)** 13 kWh

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. List two details that matter and two that can be ignored when modelling a cricket ball hit for a six. 2

22. 9 g of water is decomposed. What are the masses of hydrogen and oxygen obtained? 2

23. A boy walks 3 m east and then 4 m north. Find the distance and the magnitude of displacement. 2

24. Why do action and reaction not cancel each other? 2

25. Why does a plant cell not burst in pure water while an animal cell may? 2

26. Define a tissue. Why is division of labour useful? 2

27. What are chromosomes? How many are in a human body cell? 2

28. Name the four spheres of the Earth system. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. The mixture of an immiscible liquid pair of the same density forms no layers. Explain. 3

30. Why is it said that electrons do not move in well-defined orbits as in Bohr's model? 3

31. Differentiate between potential and kinetic energy with an example each. 3

32. Explain the working of the human ear in brief. 3

33. How does the rib cage help in breathing? 3

34. State three features of annelids. 3

35. Show the formation of an oxygen molecule. What type of bond is formed? 3

OR

Sodium chloride has Na : Cl = 23 : 35.5 by mass. How much chlorine reacts with 46 g of sodium? How much NaCl is formed? 3

36. Explain how the ozone hole formed and how it is recovering. 3

OR

Why is the rise in atmospheric CO_2 a concern? Mention two effects. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A student compares sodium chloride and sugar. 5

(i) Which conducts electricity in aqueous solution?
(ii) Why?
(iii) Compare their melting points.
(iv) Classify them as ionic or covalent.

OR

(i) Show that the combining ratio of carbon and oxygen in CO_2 is 3 : 8. 5
(ii) How much carbon is in 22 g of CO_2?
(iii) Find the mass of oxygen needed to burn 6 g of carbon completely.

38. (i) Differentiate mass and weight. 5
(ii) What is the weight of a 60 kg person on the Earth (g = 10)?
(iii) On the Moon where g = 1.6?

OR

(i) State the three laws with an example each. 5
(ii) Which law explains why seat belts are essential?

39. (i) How does the structure of a red blood cell suit its function? 5
(ii) Give another example of a specialised cell.
(iii) What is the length of an RBC lifespan?

OR

Case study: In cell culture, cells are grown in a nutrient medium at a fixed temperature. 5
(i) What is cell culture?
(ii) Why must conditions be sterile?
(iii) Name one use.
(iv) What is totipotency?

40. (i) What is the role of meiosis in sexual reproduction? 5

(ii) How many chromosomes does a human sperm, egg and zygote have?

(iii) Explain how variation arises among siblings.

OR

Case study: ASHA workers advise mothers on care during pregnancy and after birth. 5

(i) Name one thing she suggests for the mother's diet.

(ii) Why is breast-feeding recommended?

(iii) What is post-partum depression?

(iv) Why should hygiene be maintained?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND